

2026-06-14 The first short proposal

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Outline

- 1 The story
- 2 Some issues to be discussed
- 3 Related literature

What is the best part of LinkedIn data

With Revelio lab data, we know the lifecycle of a vacancy.

- From job posting data, we have information on the birth of a vacancy.
 - Full job advertisements in text: Description and requirements of the job.
 - Some standardized variables: Company, job title, location, salary range (if existing).
- From individual profiles data, we have information on who filled in the vacancy.
 - Education, gender, race (estimated), location, etc.
 - Past and future work experiences: In particular, subsequent career progression (promotion, exit, occupation transition, etc).

The research question: Different hiring strategies

Consider a sector where asymmetric information matters. In reality, firms could use different practices to decide who will fill in a vacancy. At two extremes:

- Screening strategy:
 - High job requirements, more tests and interviews. The aim is to improve the average productivity (and reduce the variance) of the new hires.
- Hiring-first strategy:
 - The firm may spend fewer resources to screen workers before hiring (due to, e.g., high screening costs, low screening precision). Instead, through employer learning on the job, the firm may fire unqualified employees with low productivity and promote only those with high productivity.
- These two extreme cases have different implications on employees' future career paths, and job advertisements will have different patterns. In reality, firms will employ a mix of these two extreme strategies.
- *The research question is: with data on workers' career paths and description of vacancies, can we uncover meaningful patterns to describe unobservable hiring practices?*

A data-driven method to classify firms by their hiring practices

This will be the biggest contribution to the literature and yet the most difficult part.

- Contributions:
 - We only know what kind of people a vacancy wants, who actually fills in the vacancy and their subsequent career trajectories. Even though we don't know how firms actually make the hiring decisions, we hope to say something about it in a data-driven way.
 - Another implication is when firms hire, they may also take future firing decisions into consideration.
- The rough idea is as follows.
 - We construct a series of variables from job posting and individual profiles (some textual analysis is useful).
 - Then we classify firms into different clusters with some data-driven methods (essentially an unsupervised machine learning problem).
 - Even though we don't know firms' real hiring practices, hopefully, we can open this black box through analysis on the creation and filling of vacancies.

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- Difficulties:
 - Suppose we have constructed a dataset about a universe of firms hiring the same occupation over years 2010-18. Each firm has multiple vacancies over the years; in each vacancy, multiple employees join the firm; for each employee, we observe his career path.
 - Difficulty 1: *We need to decide which variables to construct (and how to construct them from text) with these two sources.*
 - Difficulty 2: *We need to design an algorithm to aggregate these variables at different levels to a firm-level variable denoting their hiring practices.*

Validation and interpretation of our measure

- The hiring strategy of a firm is not random – it is already optimized based on its market conditions.
 - In particular, the productivity distribution of potential applicants and firing costs will be important factors to shape a firm's hiring strategy.
 - We may expect firms differ systematically in their hiring strategies in certain ways.
- How to interpret our measure?
 - Does it measure firms' hiring practices or does it measure firms' training practices?
 - Because we try to use workers' subsequent career paths (or some external productivity measures) to decide who are hired to fill in a particular vacancy, this may also reflect firms' training (general vs firm-specific human capital), culture, etc.

Implications of our measure

Classification $\implies$ Validation and interpretation $\implies$ Implications

- What are the implications if firms have different hiring strategies?
- We could collect external objective productivity measures (e.g., patents, clinical trials results) for firms or ideally, for individuals.
- With this productivity measure, we could investigate how firms' hiring strategies affect (1) sorting patterns between employees and firms; (2) firm productivity; (3) economy-wide productivity and efficiency.

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The occupation(s) of interest

- My current story at mind is to describe different firms within one sector (i.e., hiring one occupation/job).¹
- The first big decision is to decide which occupations we are investigating.
- Two criteria:
 - Asymmetric information matters: Entry-level jobs instead of senior hires (where more signals on productivity are available).
 - Available objective productivity measures: Patents, lawsuits results, clinical trials, etc.
- The best approach may be to (1) first pick some good candidates occupations², then (2) investigate the coverage patterns in LinkedIn data and the presence of objective productivity measures.

¹Of course, we may expect firms in different sectors use very different hiring practices (due to, e.g., different screening costs and precision). One concern is whether there is enough dispersion in firms' hiring practices within one sector.

²Some anecdotal evidence about how different firms in the same sector hire workers will be super helpful.

Another story from the worker side

The main goal is to combine two sources of data to fully exploit the advantages of LinkedIn data: job posting + individual profiles.

- In short, the above story is from firms' perspectives – how firms hire workers.
- Another potential story is from workers' perspectives:
 - Suppose for each worker, we know his full career paths, and the corresponding job advertisements for each job he has experienced.
 - Then we treat job advertisements as a detailed description of a worker's skill set (e.g., his first job requires Excel, while his second job requires Python, R).
 - A worker filling in that vacancy must indicate he has developed that skill.
 - Then we can know the lifecycle of skill upgrading for workers (e.g., from less technical to more technical; from cognitive to more social skills).

Next steps

- For the LinkedIn data, some descriptive results about (1) coverage patterns for the occupation(s) of interest; (2) the matching from individual profiles to job posting (so that we know who fills in the vacancy); (3) job advertisements text.
- For the external productivity data, some background information about (1) the sources and context; (2) how to link them to individuals or firms.

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Hiring practices

- Two chapters in the *Handbook of Labor Economics*: Hoffman and Stanton, 2025; Paul and Scott, 2011.
- Specific hiring practices/technologies:
 - Hiring as a bandit problem (exploration vs exploitation): Lazear, 1998; Li et al., 2026.
 - Referrals: Friebe et al., 2023; Hensvik and Skans, 2016; Pallais and Sands, 2016.
 - Screening tests: Autor and Scarborough, 2008; Hoffman, Kahn, et al., 2018; Mocanu, 2024.
 - Signals/Information: Agan et al., 2025; Bryan et al., 2026; Stanton and Thomas, 2016
- Not so much research (if any) on different hiring practices and their consequences across firms.

Data-driven method to uncover unobserved heterogeneity and practices

The following papers are incomplete. We need to do a comprehensive literature review on this part.

- **Econometrics:** Bonhomme, Lamadon, et al., 2022; Bonhomme and Manresa, 2015
- **Classifying individual heterogeneity:** Ahn et al., 2024; Castro et al., 2026; Gregory et al., 2025; Shibata, 2019
- **(Dis)Aggregated worker flows to describe labor markets:** Ewens and Giroud, 2025; Huitfeldt et al., 2023; Sorkin, 2018

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④ Short summary of relevant papers

Classifying individual heterogeneity

Describing labor markets

⑤ References

K-means to classify individuals based on their pattern of transitions across employment states

Gregory et al., 2025: The Alpha Beta Gamma of the Labor Market; Journal of Monetary Economics; 2025

- Data: Longitudinal Employer-Household Dynamics (LEHD); 1997-2014; the history of employment transitions for over 500,000 individual workers.
- Variables: The time spent in unemployment, the distribution of duration of different unemployment spells, and the distribution of duration of different jobs.
- The algorithm: K-means for 3 types of workers.
 - α : These workers are most likely to move from unemployment to employment within a quarter and, once they become employed, they are most likely to keep their job for more than 2 years.
 - γ : These workers are most likely to remain unemployed for more than 1 year and, once they become employed, they are most likely to leave their job within 1 year.
 - β : The pattern of employment transitions for β s is between the pattern for α s and γ s.

K-means to classify individuals based on their pattern of transitions across employment states

Castro et al., 2026: Predicting Labor Force Types; NBER Working Paper Series; 2026

- Data: NLSY79.
- Variables: Average durations of employment and non-employment spells; fraction of weeks spent out of the labor force or unemployed; number of jobs relative to the number of quarters employed.
- The algorithm: K-means to classify individuals as low- and high-attachment types.
 - Using only variables in the prime age (30-50), they first cluster individuals into low- and high-attachment types.
 - Then, they study if early-adulthood variables (before age 30, which are not used in clustering) can predict an individual's type.

HMM to classify individuals into different hidden states

Ahn et al., 2024: The Dual U.S. Labor Market Uncovered; Working Paper; 2024³

- Data: CPS 1980-2021.
- The algorithm: Hidden Markov Model (HMM).
 - 12 hidden states: 3 segments of the economy (primary, secondary, tertiary) $\times$ 4 types of employment states (employed; short-term unemployed; long-term unemployed; non-participants).
 - Observed variables (“emissions” in the terminology of HMM): Coarse classification of employment status (employed, unemployment, or not-in-the-labor-force).
- Unlike the previous papers where an individual type is fixed, in this paper, classifications of individual-month cells into 12 hidden states depend on transition histories (which is modelled as a markov process).

³A similar paper is Shibata, 2019: “Labor Market Dynamics: A Hidden Markov Approach; Working Paper (Conditionally Accepted at Journal of the European Economic Association); 2019”.

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A worker flow approach to internal labor markets

Huitfeldt et al., 2023: Internal labor markets: A worker flow approach; Journal of Econometrics; 2023

- Data: Norway matched employer-employee data (restricted to large firms), 3,611 firms, with 7-digit occupational codes (i.e., job titles).
- The algorithm: A data-driven two-step estimator combining economic theory and clustering.
 - Step 1: Use worker flows to identify different internal labor markets (ILM) within a firm. An ILM comprises occupations within the firm that are connected by some path of realized worker flows.⁴
 - Step 2: Rank jobs within ILMs using relative flows between them⁵, then cluster occupations with similar ranks into the same level of hierarchy with a k-means clustering algorithm⁶.

⁴Consider an intuitive example firm in the manufacturing sector, where the majority of production workers are part of a large ILM, including production line management positions and top executives. Within the ILM, workers switch positions, e.g., through promotions or job rotation. In a separate ILM, the firm employs workers for logistical purposes, like security personnel or receptionists. There are, however, no worker flows from one ILM to the other one – the two ILMs are entirely unconnected. This basic idea is captured in the concept of connected components from graph theory.

⁵The basic intuition is simple: more observed transitions from occupation b to occupation a than vice-versa imply that occupation a is ranked higher than b .

⁶A layer in a firm is a collection of jobs that are more likely to have lateral moves between them rather than promotion or demotion

A worker flow approach to internal labor markets

Ewens and Giroud, 2025: Corporate Hierarchy; NBER Working Paper Series; 2025

- Data: LinkedIn data (worker resumes); Publicly-traded firms in USA
- The algorithm: Apply the Huitfeldt et al., 2023 two-step estimator to have a firm-year measure of the number of hierarchical layers.

PageRank to rank firms with employer-to-employer transitions

Sorkin, 2018: Ranking Firms Using Revealed Preference; The Quarterly Journal of Economics; 2018

- Data: Longitudinal Employer Household Dynamics (LEHD) data; Transitions (quarterly) of employer-to-employer or employer-to-nonemployment-to-employer (ENE).
- The algorithm: Google's PageRank.⁷
- The intuition:
 - Suppose there are two firms: A and B. Suppose that both firms are initially the same size and make the same number of offers to workers at the other firm at random.
 - If more workers accept A's offer than accept B's offer, then we can infer that workers prefer firm A to firm B. If it also turns out that B is higher-paying than A, then we infer that B offers worse nonpay characteristics than A (since workers prefer A to B despite the lower pay). Hence, compensating differentials explains why B pays more than A.

⁷A recursive definition of good: "good firms hire from other good firms and have few workers leave."

Outline

4 Short summary of relevant papers

5 References

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